EDA for the road features extracted from images and contextual database

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EDA Report: Dataset (model_dataset 1.xlsx)

Introduction

This report presents an exploratory data analysis (EDA) of `model_dataset 1.xlsx`, a dataset of 49 road segments in Western Australia. The contextual variables (e.g. population statistics, distance to public transport, health, and education facilities, building and retail counts) were extracted from a contextual ecosocial database, while the image-derived variables (e.g. footpath type, land use, and the presence of nearby amenities) were obtained from earlier feature extraction work. The goal is to identify variables that are associated with observed pedestrian counts, in preparation for building a predictive regression model.

Dataset and Variables

The dataset contains two response variables, measuring pedestrian counts per hour in two directions:

- `obs_peak_along_ped_per_h`
- `obs_peak_cross_ped_per_h`

The remaining variables fall into two groups: 21 numeric variables (e.g. building counts, population density, distance to public transport, health, and education facilities) and 16 categorical variables (e.g. footpath type, land use, and the presence of nearby amenities).

Table 1: Dataset: variable types and unique value counts

variable	type	n_unique
obs_peak_cross_ped_per_h	numeric	39
obs_peak_along_ped_per_h	numeric	42
abs_mb_area_sqkm_2021	numeric	48
abs_mb_dwellings_2021	numeric	23
abs_mb_persons_usual_resident_2021	numeric	26
abs_mb_dwellings_per_sqkm_2021	numeric	30
abs_mb_persons_per_sqkm_2021	numeric	30
pta_nearest_active_stop_distance_m	numeric	49
pta_active_stops_within_500m	numeric	15
health_nearest_establishment_distance_m	numeric	49
education_nearest_active_school_distance_m	numeric	49
ms_building_count_500m	numeric	47
ms_building_count_per_sqkm_500m	numeric	47
ms_building_footprint_area_m2_500m	numeric	49
ms_building_coverage_ratio_500m	numeric	49
overture_food_drink_count_500m	numeric	17

variable	type	n_unique
overture_grocery_convenience_pharmacy_count_500m	numeric	10
overture_other_shopping_count_500m	numeric	21
pedestrian_crossings	numeric	3
commercial_buildings	numeric	12
residential_buildings	numeric	13
industrial_buildings	numeric	4
parking_areas	numeric	5
abs_mb_category_2021	character	7
pta_nearest_active_stop_type	character	3
health_nearest_establishment_category	character	2
health_nearest_establishment_type	character	2
education_nearest_school_sector	character	2
education_nearest_school_low_year	character	5
education_nearest_school_high_year	character	3
overture_shopping_mall_present_500m	logical	2
footpath_both_sides	logical	2
footpath_one_side	logical	2
fencing_barriers	logical	2
railway_station	logical	2
bus_stops	logical	2
overpass_bridge	logical	2
park_bushland_reserve	logical	2
shared_pedestrian_and_cycle_path	logical	2

Numeric Variables

Correlation coefficients were calculated between each numeric variable and both response variables.

Table 2: Correlation coefficients between numeric variables and response variables

	obs_peak_cross_ped_per_h	obs_peak_along_ped_per_h
abs_mb_area_sqkm_2021	-0.15	-0.13
abs_mb_dwellings_2021	-0.05	-0.07
abs_mb_persons_usual_resident_2021	0.05	-0.03
abs_mb_dwellings_per_sqkm_2021	-0.12	-0.09
abs_mb_persons_per_sqkm_2021	-0.01	-0.05
pta_nearest_active_stop_distance_m	0.16	-0.08
pta_active_stops_within_500m	-0.06	0.02
health_nearest_establishment_distance_m	0.00	-0.01
education_nearest_active_school_distance_m	-0.15	-0.10
ms_building_count_500m	0.08	0.09
ms_building_count_per_sqkm_500m	0.08	0.09
ms_building_footprint_area_m2_500m	0.10	0.12
ms_building_coverage_ratio_500m	0.10	0.12
overture_food_drink_count_500m	0.57	0.59
overture_grocery_convenience_pharmacy_count_500m	0.39	0.42
overture_other_shopping_count_500m	0.29	0.42
pedestrian_crossings	0.23	0.18
commercial_buildings	0.66	0.77
residential_buildings	-0.21	-0.14

	obs_peak_cross_ped_per_h	obs_peak_along_ped_per_h
industrial_buildings	-0.08	-0.08
parking_areas	0.42	0.46

Findings:

Three variables show a clear correlation (|r| >= 0.4) with both response variables: `commercial_buildings` (r = 0.66–0.77), `overture_food_drink_count_500m` (r = 0.57–0.59), and `parking_areas` (r = 0.42–0.46). All three are statistically significant ($p < 0.01$) and relate to commercial and retail activity near the site.

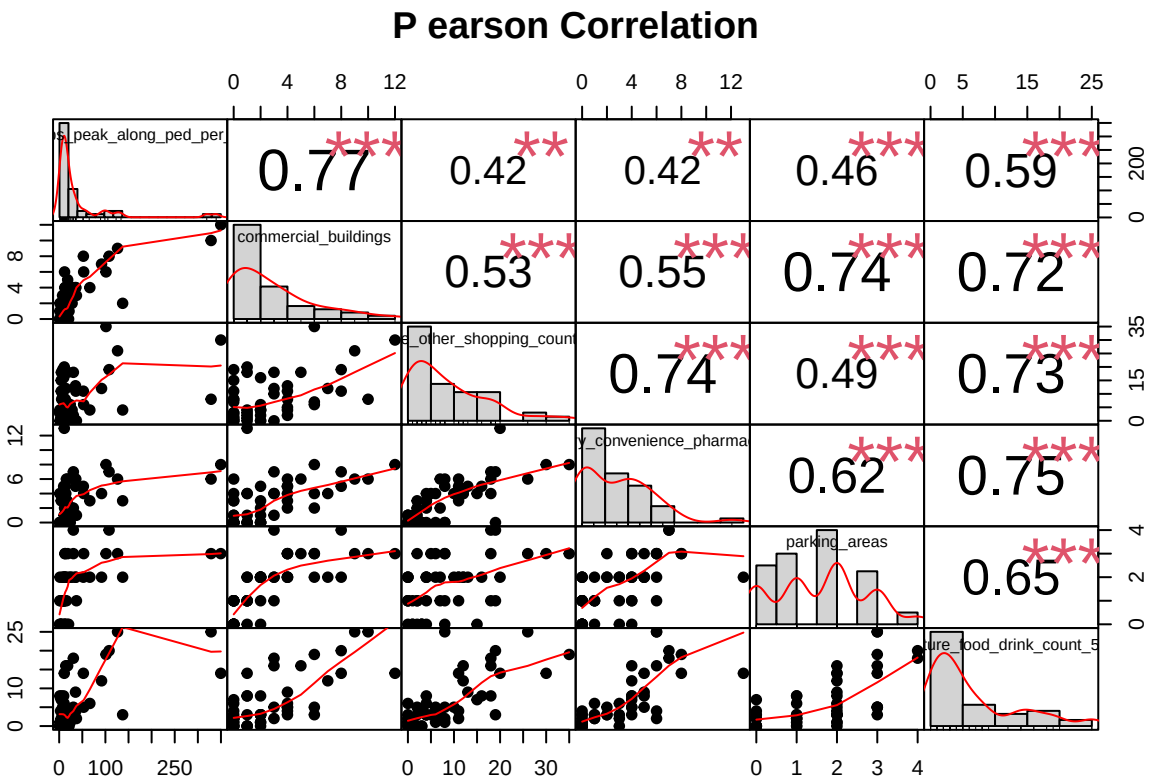
Two further retail-related variables, `overture_grocery_convenience_pharmacy_count_500m` and `overture_other_shopping_count_500m`, show a moderate correlation with `obs_peak_along_ped_per_h` (r = 0.42 for both) but a weaker correlation with `obs_peak_cross_ped_per_h` (r = 0.39 and 0.29 respectively).

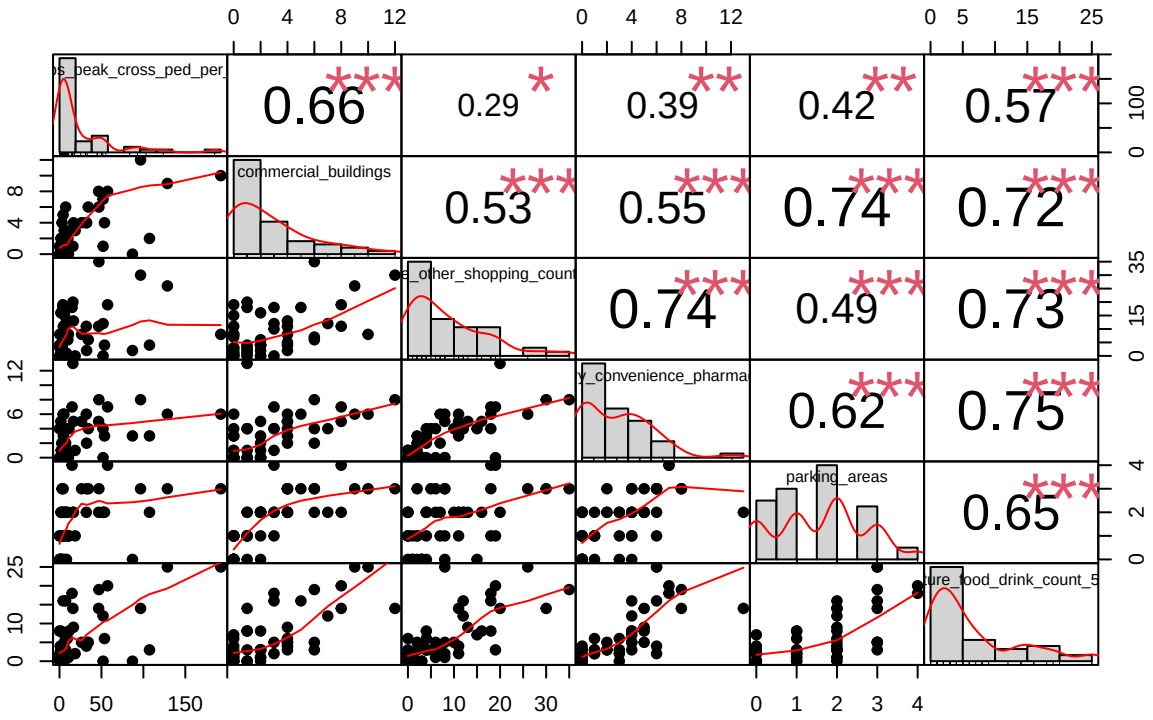
`pedestrian_crossings` and `residential_buildings` show weak correlations (|r| between 0.14 and 0.23) that are not statistically significant.

The remaining variables, including ABS population statistics, distance to public transport, health, and education facilities, and Microsoft building footprint measures, show little to no correlation (|r| < 0.2) with either target.

Five Significant Predictors

The five numeric variables identified as significant predictors of pedestrian counts — `commercial_buildings`, `overture_other_shopping_count_500m`, `overture_grocery_convenience_pharmacy_count_500m`, `parking_areas`, and `overture_food_drink_count_500m` — are examined together using scatterplot matrices, to check both their relationship with each response variable and their correlation with one another.





The five significant predictors are also highly correlated with each other (r = 0.56 to 0.87). This indicates strong multicollinearity, which will need to be addressed before these variables can be used together in a regression model.

Logical Categorical Variables

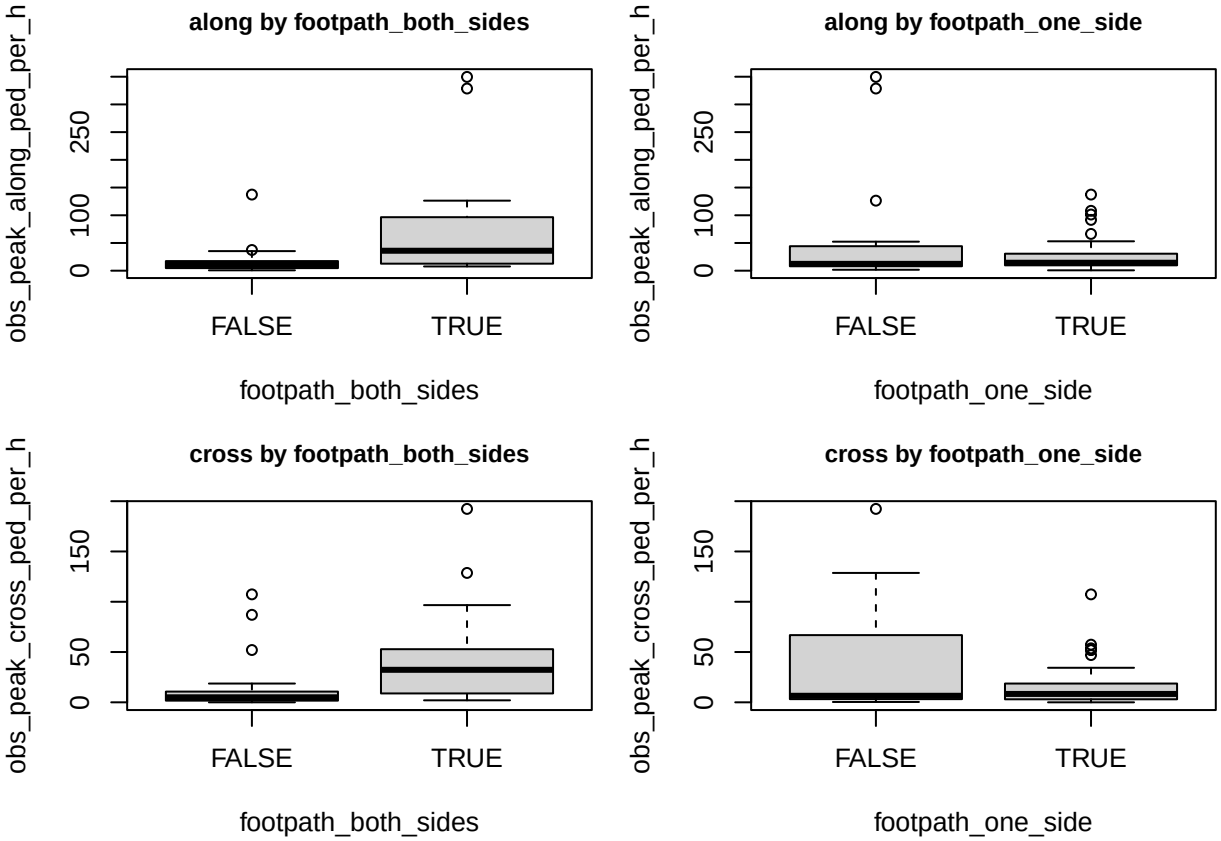
The number of TRUE and FALSE cases was checked for each logical variable, to identify variables with insufficient sample size in either group.

Table 3: TRUE/FALSE counts for logical variables (n=49)

variable	FALSE_n	TRUE_n
overture_shopping_mall_present_500m	40	9
footpath_both_sides	30	19
footpath_one_side	16	33
fencing_barriers	3	46
railway_station	47	2
bus_stops	47	2
overpass_bridge	48	1
park_bushland_reserve	8	41
shared_pedestrian_and_cycle_path	36	13

The table shows a wide range of balance across variables. `fencing_barriers`, `railway_station`, `bus_stops`, and `overpass_bridge` each have fewer than 5 cases in one group, indicating a strong imbalance that limits the reliability of any comparison. The remaining variables — `overture_shopping_mall_present_500m`, `footpath_both_sides`, `footpath_one_side`, `park_bushland_reserve`, and `shared_pedestrian_and_cycle_path` — have at least 5 cases in both groups, though the degree of balance still varies considerably between them.

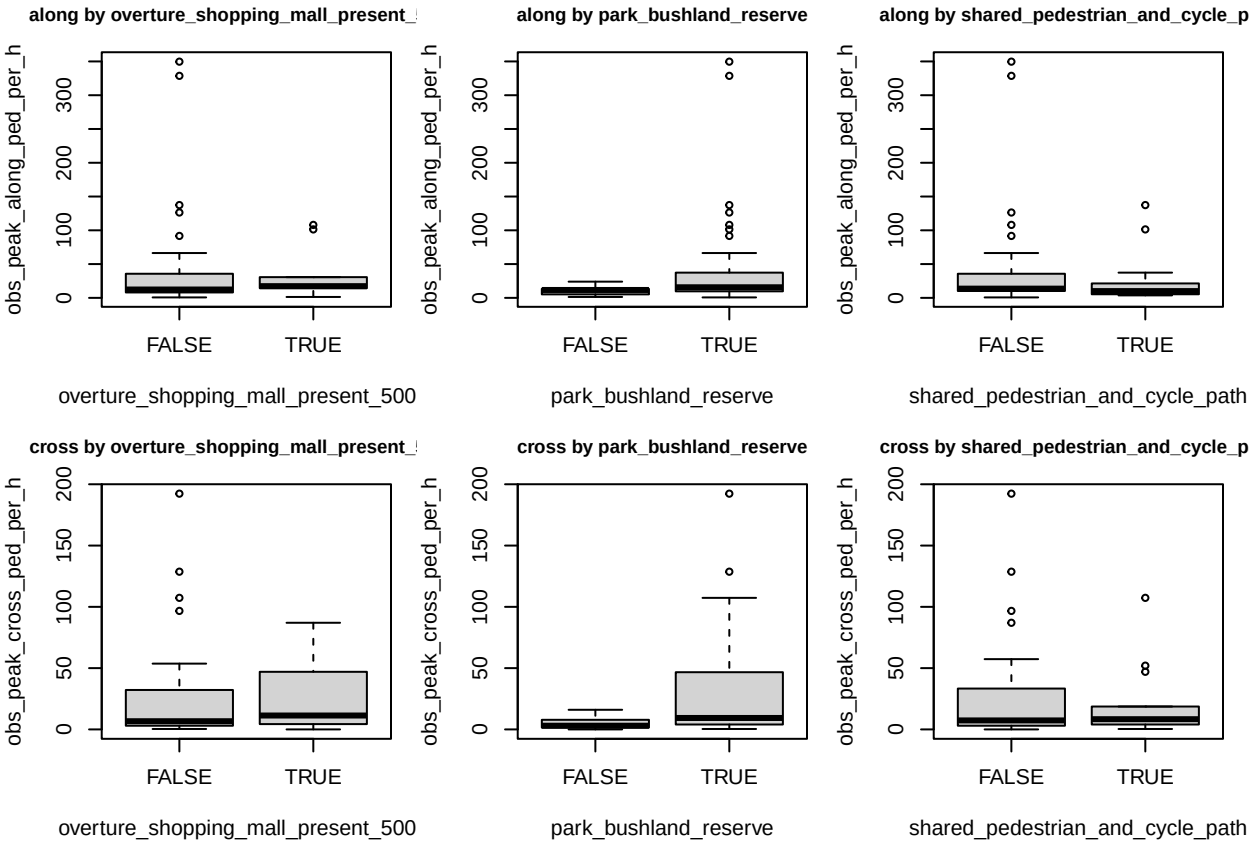
Footpath Category



`footpath_both_sides` shows a clear pattern: segments with footpaths on both sides (`TRUE`) have a higher median and wider spread for both response variables, compared to segments without (`FALSE`).

`footpath_one_side` does not show a clear increase in pedestrian counts when `TRUE`; medians and spread are similar between the two groups for both response variables.

Amenity and Land Use Category



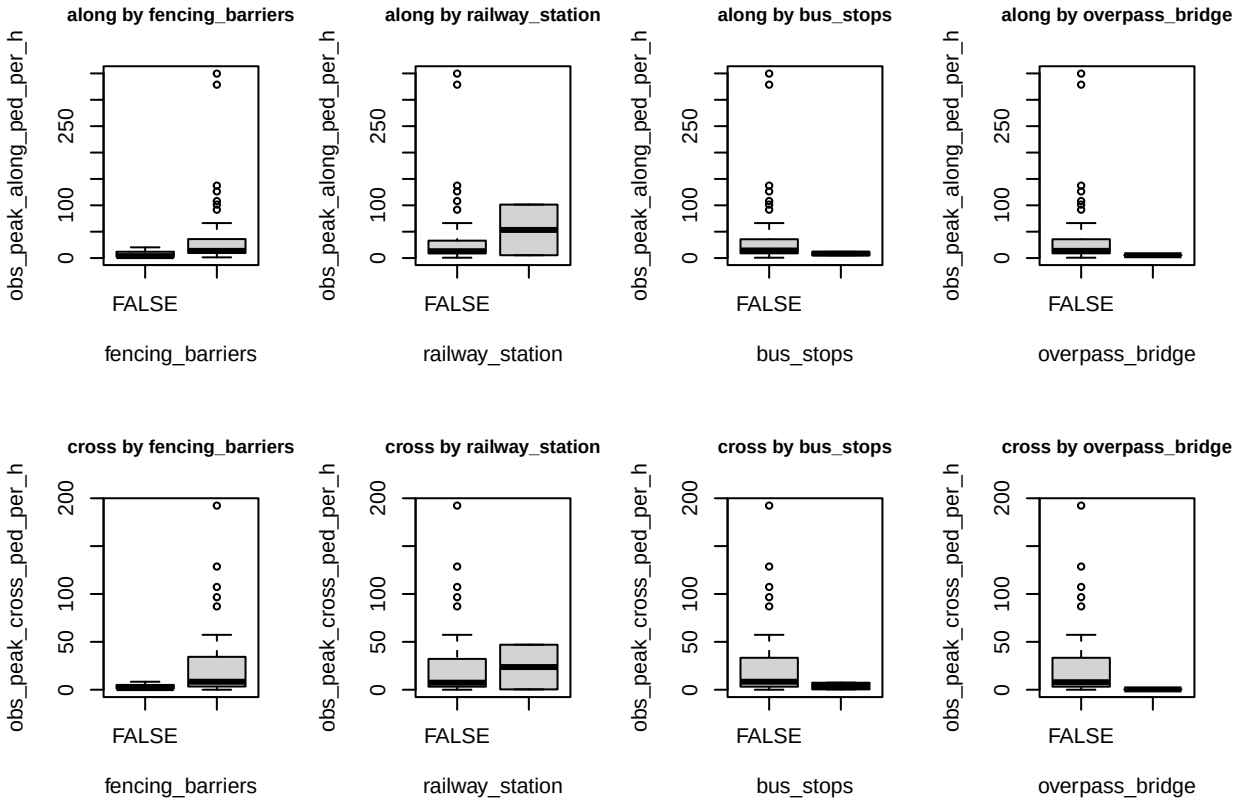
`overture_shopping_mall_present_500m` shows only a modest difference between groups: locations where a shopping mall is present within 500m (`TRUE`) have a slightly higher median than locations where it is absent, but the spread is broadly similar for both target variables.

`park_bushland_reserve` shows a similar pattern to `footpath_both_sides`: locations where this is `TRUE` have a higher median and wider spread than locations where it is `FALSE`, for both target variables.

`shared_pedestrian_and_cycle_path` shows little difference between the two groups, with overlapping boxes and similar medians for both response variables.

Excluded Small-Sample Category

As a reference, boxplots for the four excluded variables are shown below. Although the small group size (fewer than 5 cases in one group) makes formal comparisons unreliable, the plots are included to check for any obvious trend before ruling these variables out entirely.



Findings:

`fencing_barriers` and `railway_station` show a trend toward higher pedestrian counts when `TRUE`, for both response variables. However, the `FALSE` group for `fencing_barriers` (n = 3) and the `TRUE` group for `railway_station` (n = 2) are extremely small, so these apparent trends should be treated with caution rather than as reliable evidence of an effect.

`bus_stops` and `overpass_bridge` show the opposite direction, with lower pedestrian counts when `TRUE`, but the `TRUE` groups are similarly small (n = 2 and n = 1 respectively), so no reliable conclusion can be drawn.

Given the small group sizes across all four variables, none are considered suitable for further statistical analysis at this stage.

Character-Type Categorical Variables

The level distribution of each character-type categorical variable was checked.

Table 4: Level counts and percentages for character categorical variables

variable	level	n	pct
abs_mb_category_2021	Commercial	10	20.4
abs_mb_category_2021	Education	3	6.1
abs_mb_category_2021	Industrial	3	6.1
abs_mb_category_2021	Parkland	4	8.2
abs_mb_category_2021	Primary Production	1	2.0
abs_mb_category_2021	Residential	26	53.1
abs_mb_category_2021	Transport	2	4.1
pta_nearest_active_stop_type	School	10	20.4
pta_nearest_active_stop_type	Standard	11	22.4

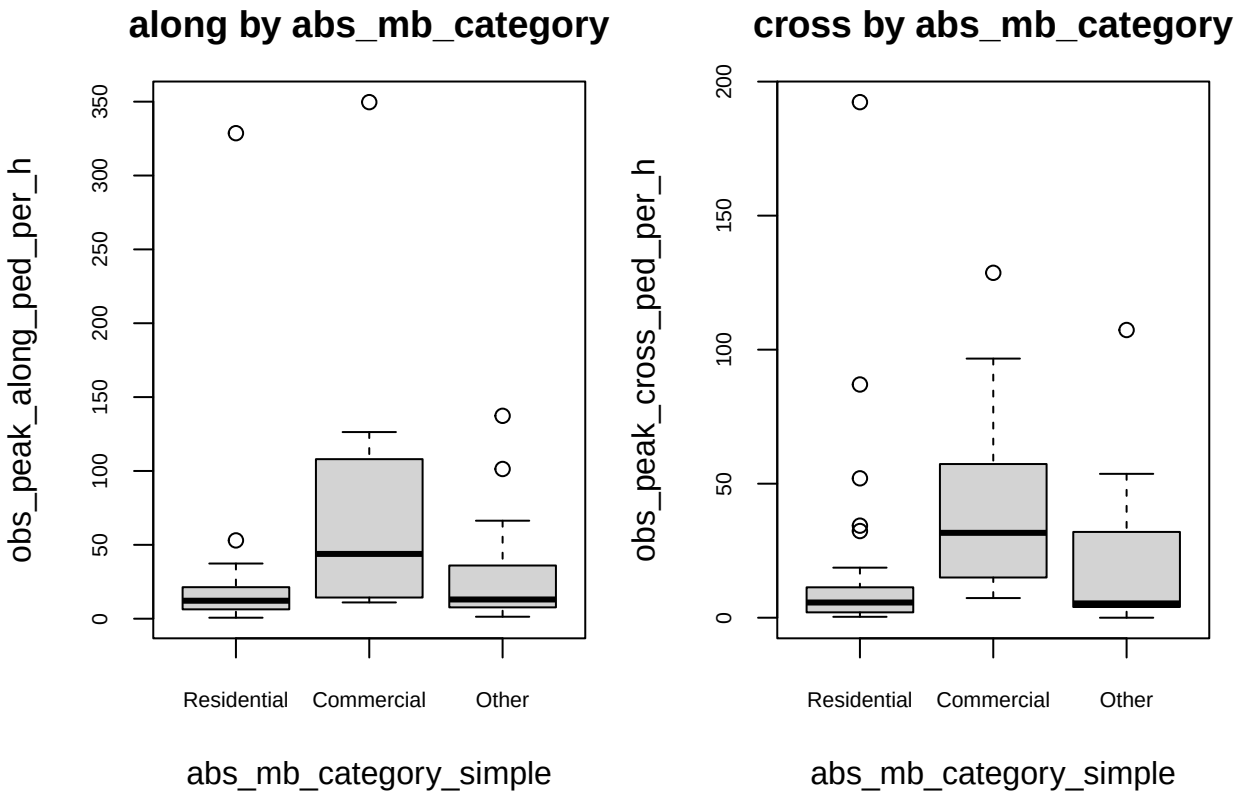
variable	level	n	pct
pta_nearest_active_stop_type	Transwa	28	57.1
health_nearest_establishment_category	Acute Hospital	44	89.8
health_nearest_establishment_category	Nursing Post	5	10.2
health_nearest_establishment_type	Private	4	8.2
health_nearest_establishment_type	Public	45	91.8
education_nearest_school_sector	GOVERNMENT	31	63.3
education_nearest_school_sector	NON-GOVERNMENT	18	36.7
education_nearest_school_low_year	KIN	35	71.4
education_nearest_school_low_year	PKG	8	16.3
education_nearest_school_low_year	Y07	4	8.2
education_nearest_school_low_year	Y10	1	2.0
education_nearest_school_low_year	Y11	1	2.0
education_nearest_school_high_year	Y06	32	65.3
education_nearest_school_high_year	Y10	4	8.2
education_nearest_school_high_year	Y12	13	26.5

The table shows considerable variation in balance across variables. `pta_nearest_active_stop_type` and `education_nearest_school_sector` are relatively well balanced across their levels. `abs_mb_category_2021` and `education_nearest_school_low_year` are each dominated by one or two levels, with several smaller levels falling below 10% of the sample. `health_nearest_establishment_category`, `health_nearest_establishment_type`, and `education_nearest_school_high_year` each include at least one level with fewer than 5 observations, limiting the reliability of comparisons involving that level. Each variable is examined in more detail in the sub-sections below.

Land Use Category

`abs_mb_category_2021` is dominated by Residential (53.1%) and Commercial (20.4%), with the remaining five levels each below 10% of the sample.

`abs_mb_category_2021` is simplified into three levels — `Residential`, `Commercial`, and `Other` (combining Education, Industrial, Parkland, Primary Production, and Transport) — since the smaller levels each account for less than 10% of the sample.

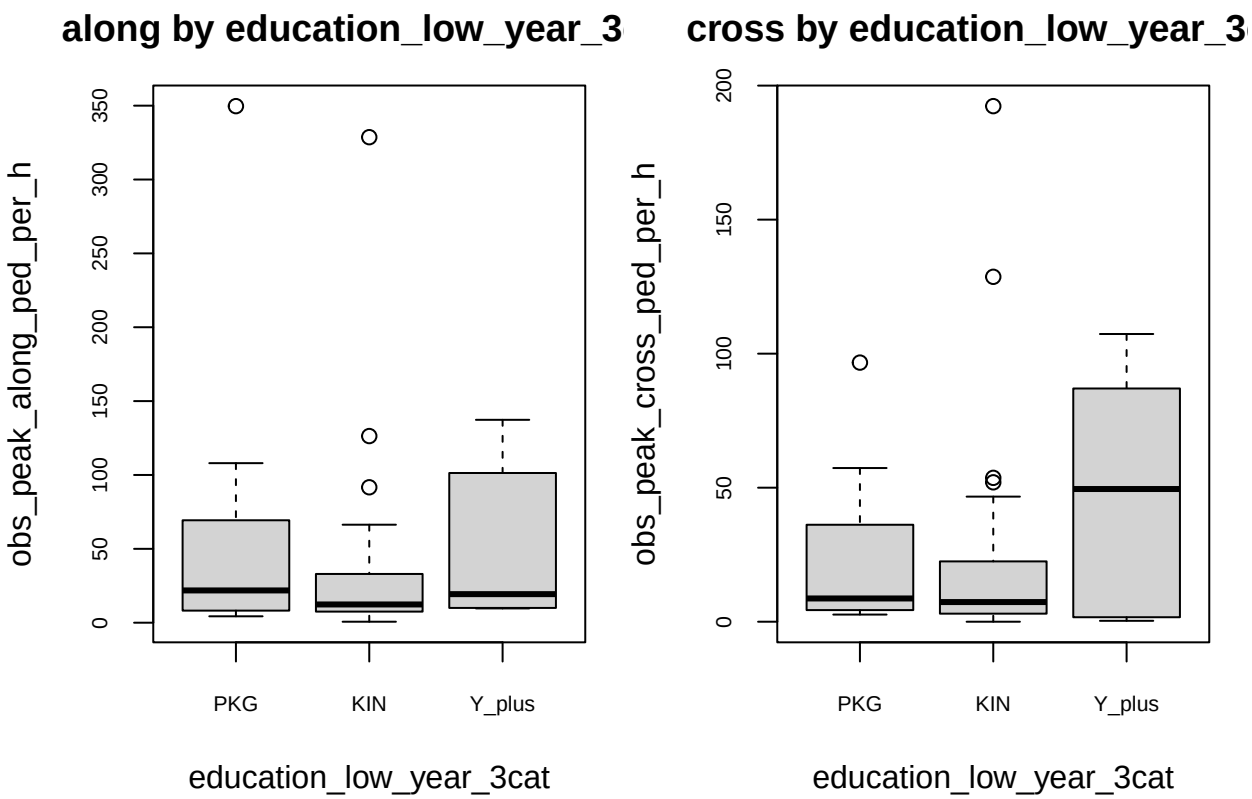


The Commercial group shows the highest median and widest spread, followed by Other, with Residential showing the lowest and narrowest distribution, for both target variables. This is consistent with the earlier finding that `commercial_buildings` was the strongest numeric predictor.

Education Category

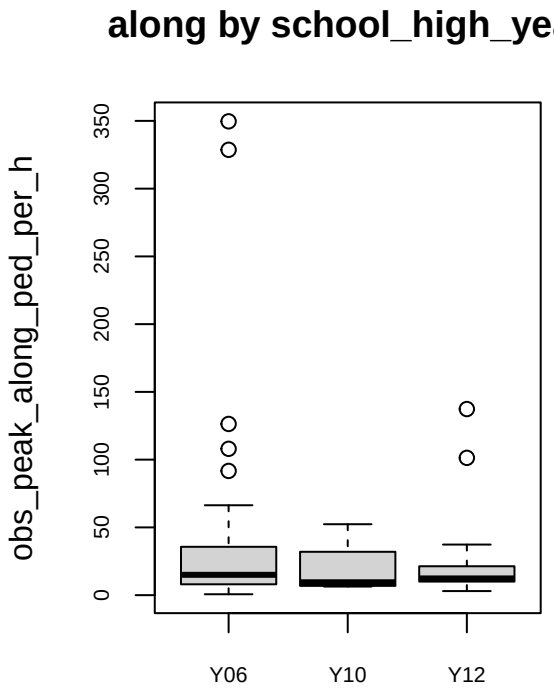
`education_nearest_school_low_year` is dominated by `KIN` (71.4%) and `PKG` (16.3%), with the remaining three levels — `Y07`, `Y10`, `Y11` — each below 10% of the sample and fewer than 5 observations.

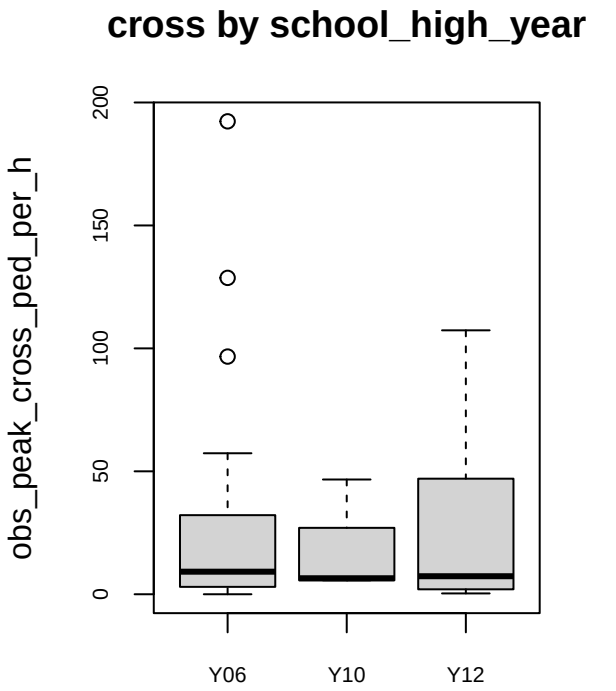
`education_nearest_school_low_year` is simplified into three levels — `PKG`, `KIN`, and `Y_plus` (combining Y07, Y10, and Y11) — since the Y07, Y10, and Y11 levels each have fewer than 5 observations.



The `Y_plus` group shows the highest median and widest spread, followed by `PKG`, with `KIN` showing the lowest and narrowest distribution, for both target variables.

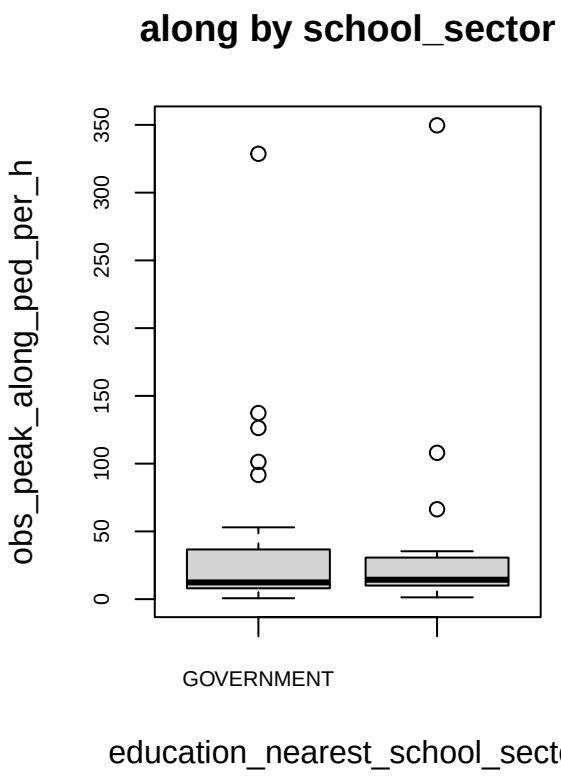
`education_nearest_school_high_year` is dominated by `Y06` (65.3%) and `Y12` (26.5%), with `Y10` accounting for 8.2% of the sample (n = 4).

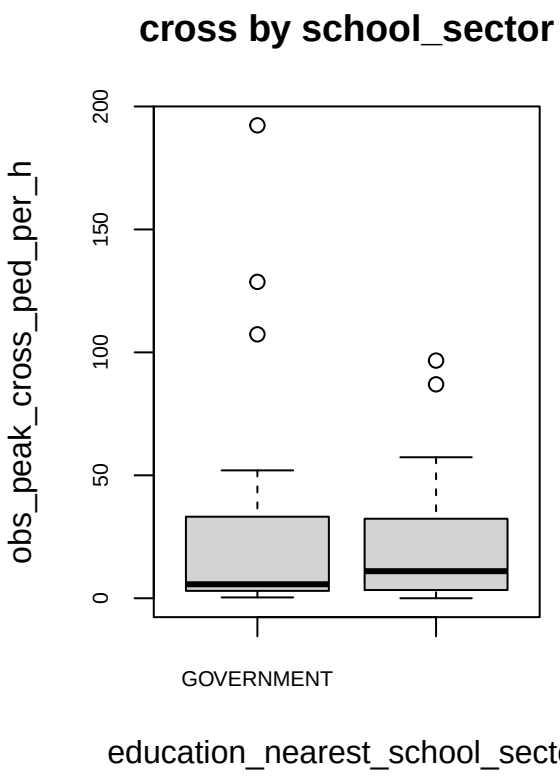




`education_nearest_school_high_year` shows no meaningful difference in pedestrian counts across the three groups, for both target variables.

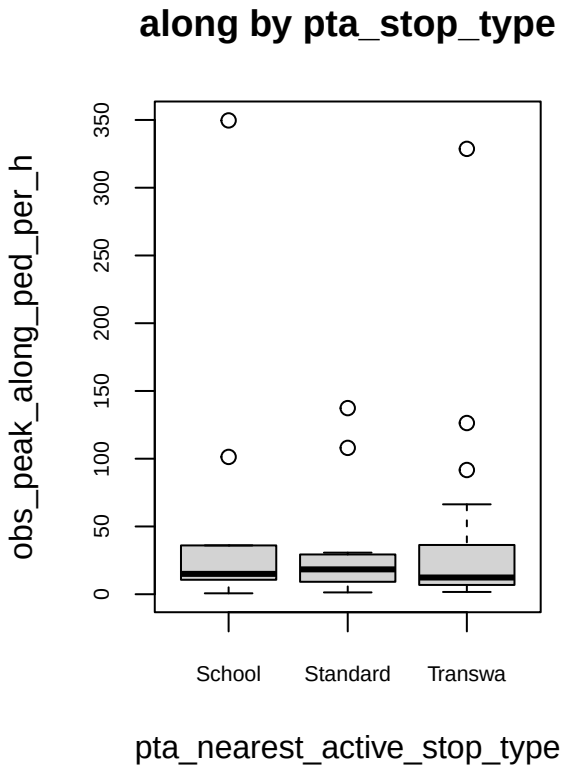
`education_nearest_school_sector` is relatively balanced between its two levels (GOVERNMENT = 63.3%, NON-GOVERNMENT = 36.7%).

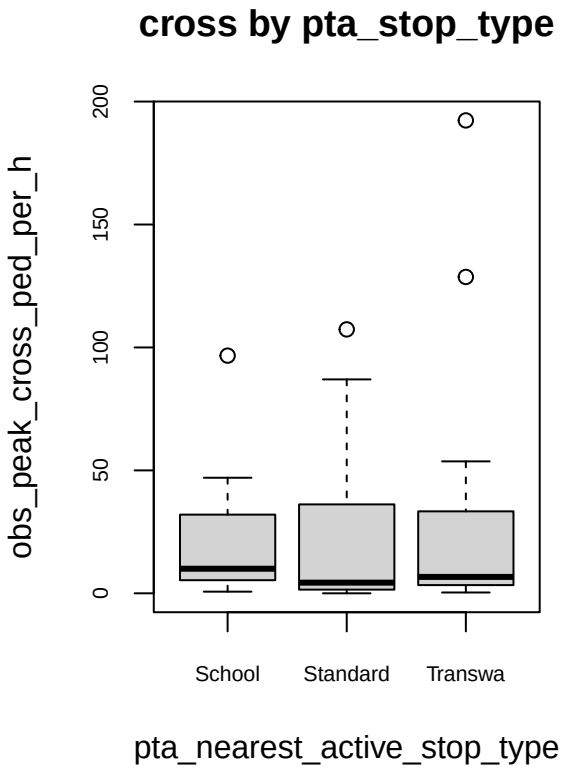




`education_nearest_school_sector` shows no meaningful difference in pedestrian counts across groups, based on these plots.

Public Transport Category

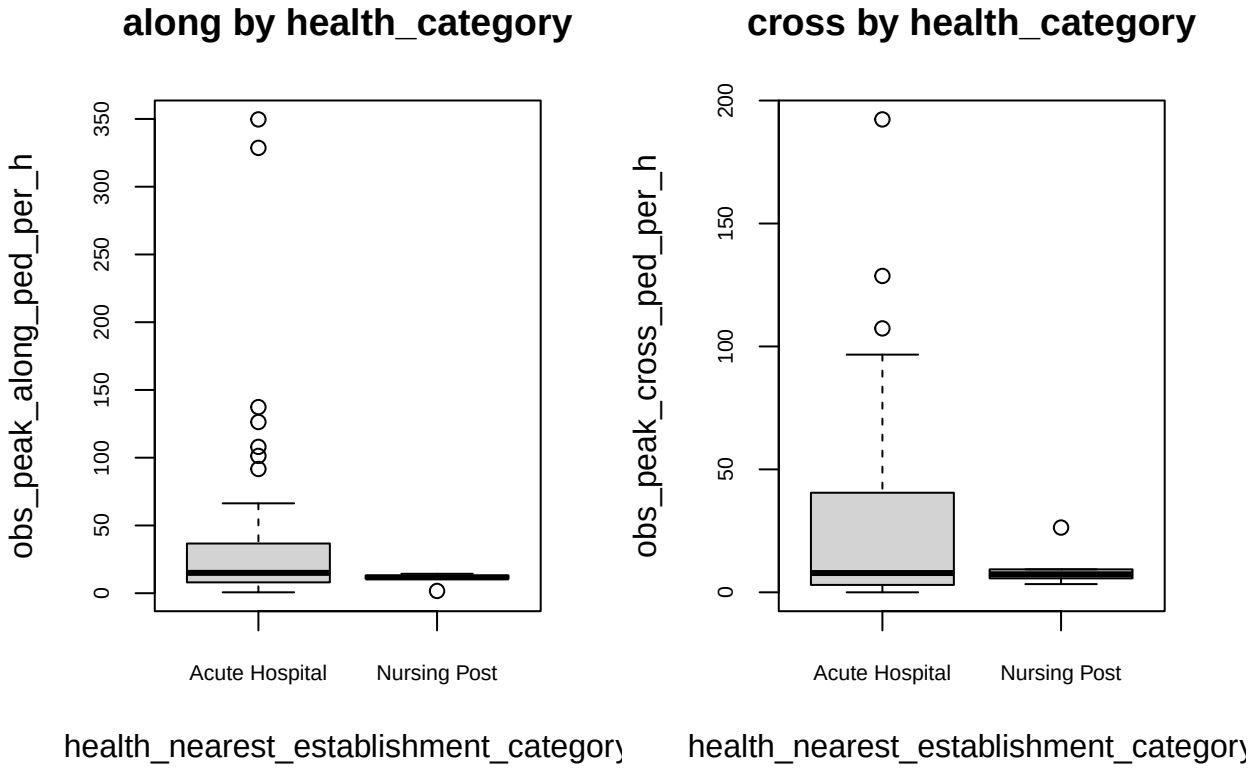




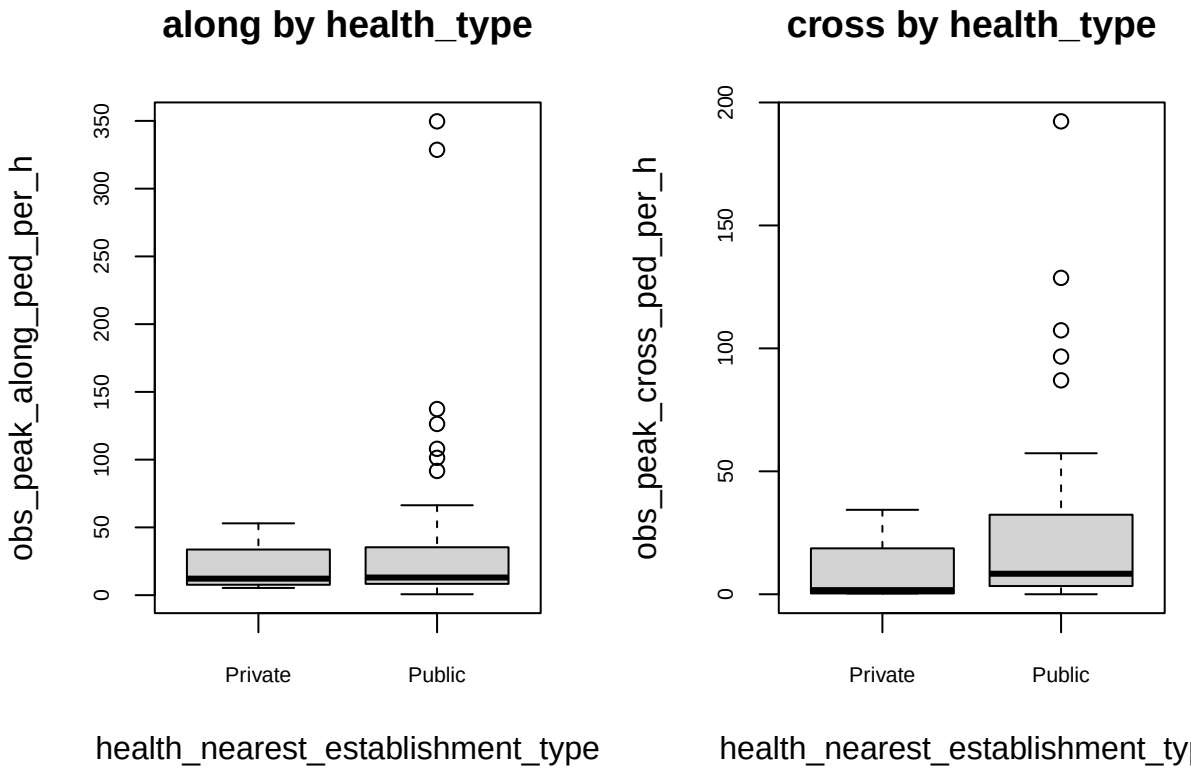
`pta_nearest_active_stop_type` shows no meaningful difference in pedestrian counts across groups, based on these plots.

Healthcare Category

As a reference, boxplots for the two health-related variables are shown below. Although the group sizes are heavily skewed (`Nursing Post` n = 5, `Private` n = 4), the plots are included to check for any obvious trend before excluding these variables from further analysis.



`health_nearest_establishment_category` shows a possible trend: sites near a `Nursing Post` have lower pedestrian counts than sites near an `Acute Hospital`, for both target variables. However, the `Nursing Post` group is very small (n = 5), so this difference is not reliable.



`health_nearest_establishment_type` shows almost no difference between `Private` and `Public` groups, for both target variables. The `Private` group is also very small (n = 4), which limits how much can be concluded.

Given the small group sizes, neither variable is considered suitable for further statistical analysis at this stage.

Conclusion

`commercial_buildings` and related retail/commercial density measures (`overture_other_shopping_count_500m`, `overture_grocery_convenience_pharmacy_count_500m`, `parking_areas`, `overture_food_drink_count_500m`) are the strongest and most consistent numeric predictors of pedestrian counts, but they are highly correlated with each other and require a multicollinearity check before joint use in a model.

Among categorical variables, `footpath_both_sides`, `overture_shopping_mall_present_500m`, `park_bushland_reserve`, land-use category (`abs_mb_category_simple`), and school entry level (`education_low_year_3cat`) show a clear relationship with pedestrian counts, all pointing toward higher counts in commercially developed, well-footpathed areas served by schools with a later entry year. `footpath_one_side`, `shared_pedestrian_and_cycle_path`, `pta_nearest_active_stop_type`, school exit level (`education_nearest_school_hig` and school sector (`education_nearest_school_sector`) show no meaningful relationship and are unlikely to add value to the model.

`health_nearest_establishment_category` and `health_nearest_establishment_type` were examined only as a reference, due to extremely small group sizes (n = 4–5) in one level of each variable, and are excluded from further consideration.

Recommended next step: examine multicollinearity among the commercial/retail density variables (e.g. via VIF or correlation-based selection) before fitting a regression model, and confirm whether `footpath_both_sides`, `abs_mb_category_simple`, and `education_low_year_3cat` add explanatory power independent of `commercial_buildings`.